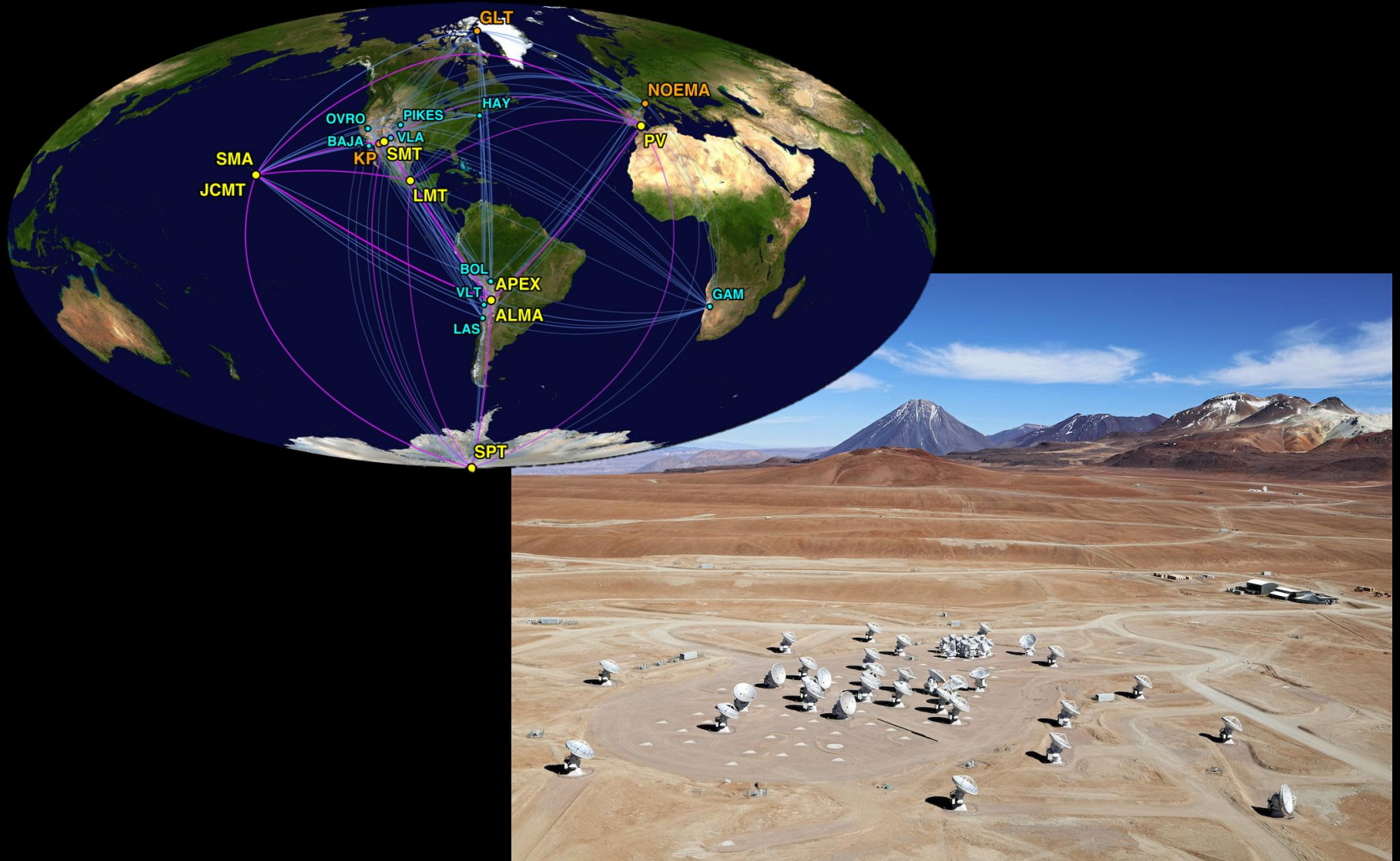




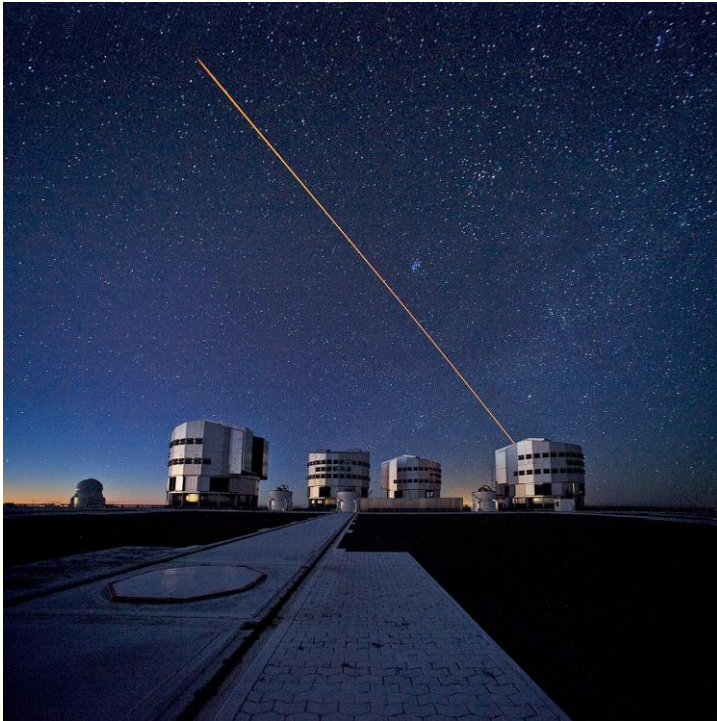
Science with the world's largest telescopes

Dr Katharine Johnston

Outline

- Big optical and radio telescopes and why we need them
- Peering through interstellar dust using radio observations
- Introducing ALMA and interferometers
- ALMA results on AFGL4176: a disk around a forming massive star (my research)
- Even bigger interferometers and the EHT
- Science results from the EHT: observations of supermassive black holes

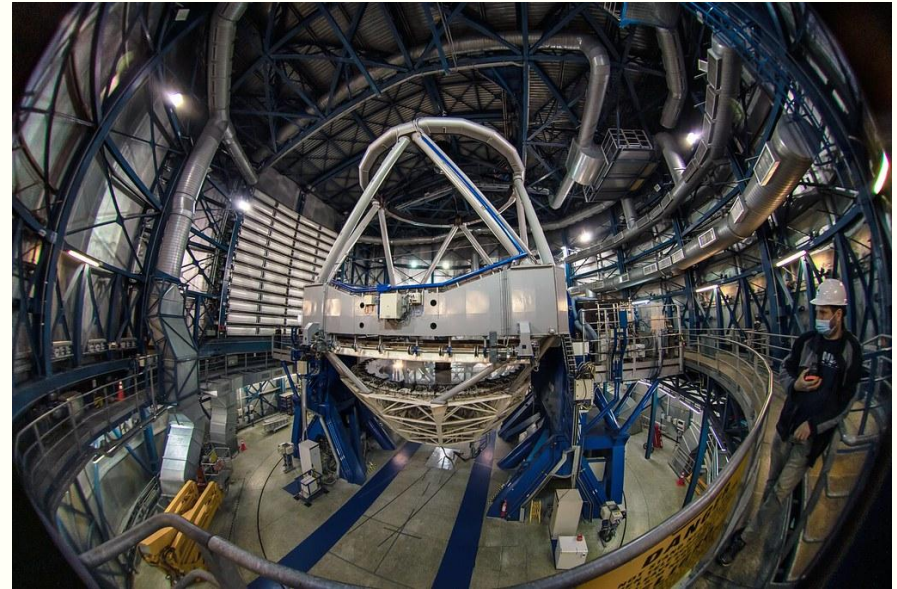
Big optical telescopes



The Very Large Telescope (VLT)
Four 8.2 m diameter telescopes

Why we need big telescopes:

1. Resolution
2. Sensitivity



Inside one of the VLT Domes

$$\theta \sim \frac{\lambda}{D}$$

Resolution (smallest detail in image) $\rightarrow \theta$

Wavelength of light $\rightarrow \lambda$

Diameter of telescope $\rightarrow D$

Big radio telescopes

Why do radio telescopes have to be bigger?

The longer the wavelength, the larger the telescope you need to observe with the same detail.

$$\theta \sim \frac{\lambda}{D}$$



FAST - Five-hundred-meter Aperture Spherical Telescope (500m diameter)



Effelsberg Telescope (100m diameter)

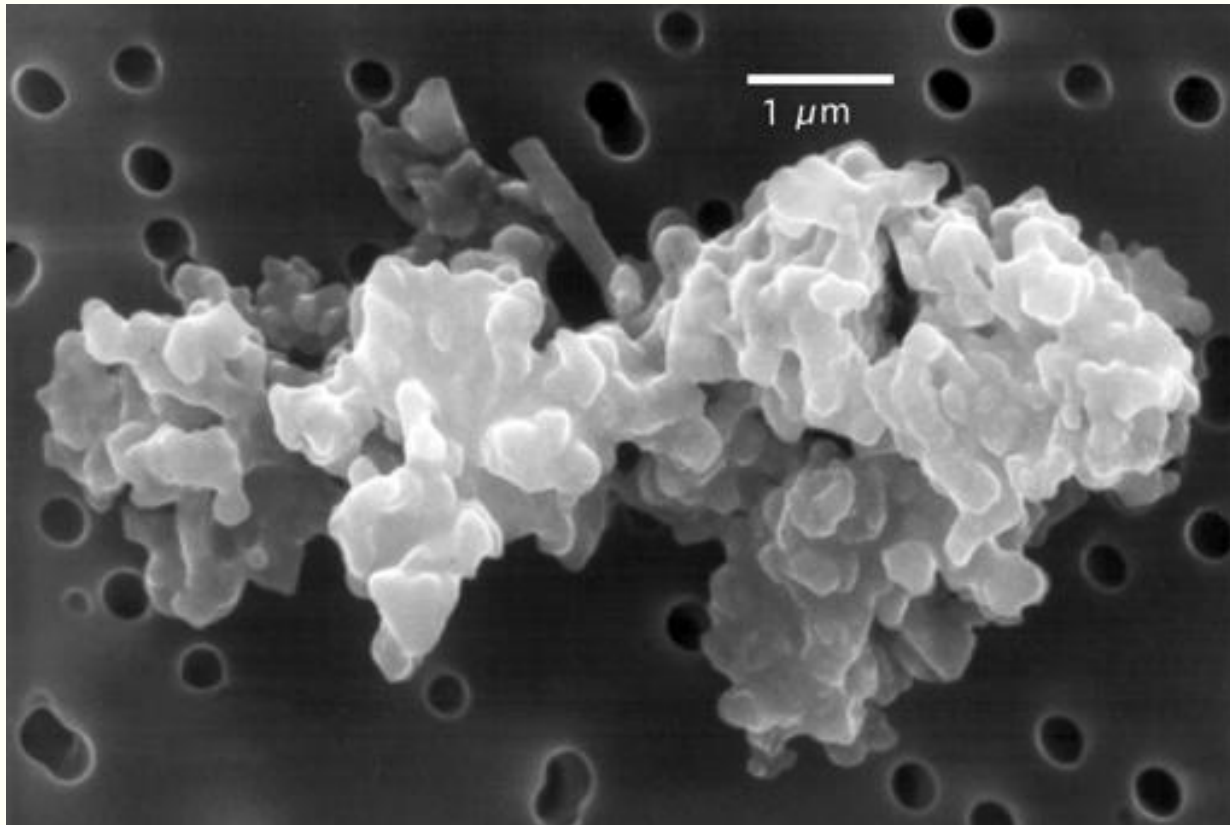


The Lovell Telescope (76m diameter)

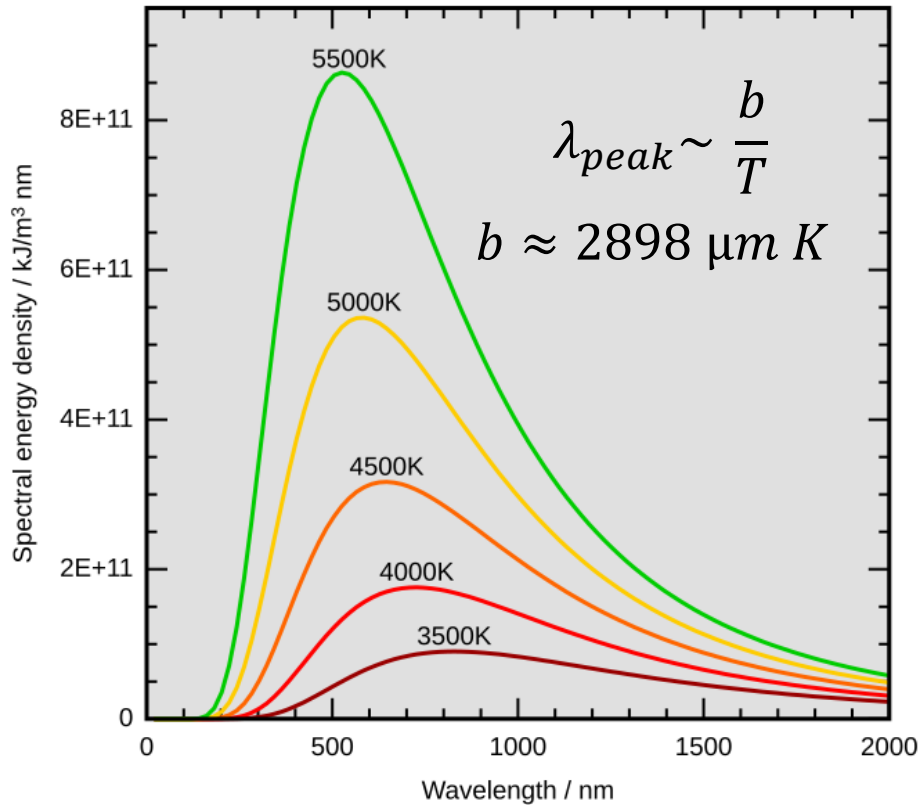


YURI
BELETSKY

The cosmic culprit: dust



Wien's law



Luminous
massive
young stars
in the
Pleiades



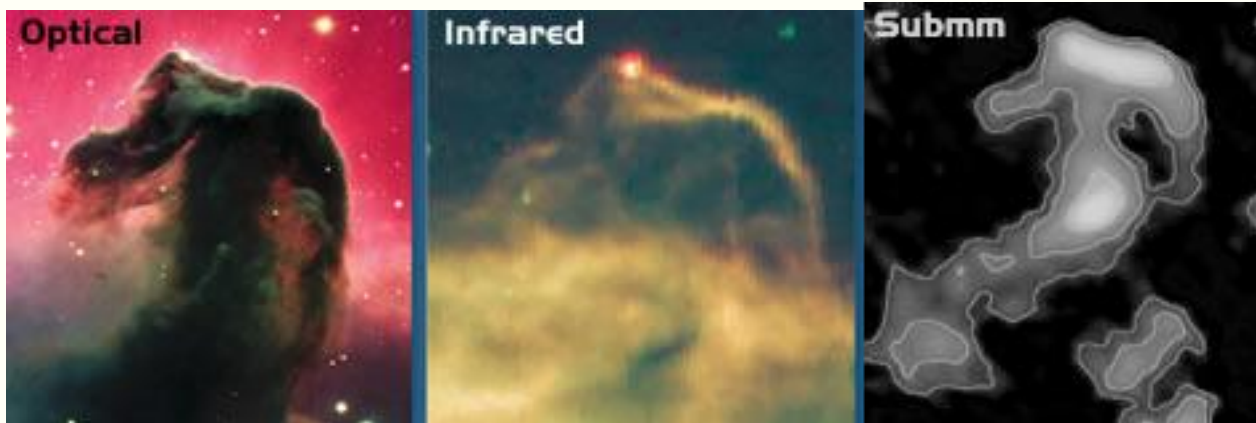
Ultracool dwarf star TRAPPIST-1
seen from the surface of one of
its planets (artist's conception)

People in the Infrared



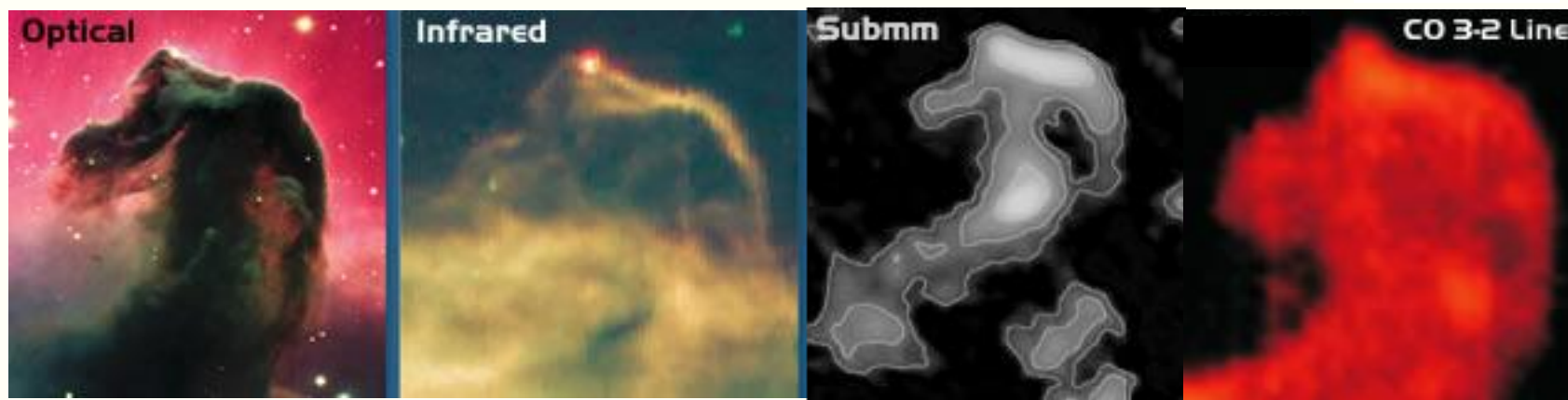
People (37°C) seen at infrared wavelengths

Peering through the dust



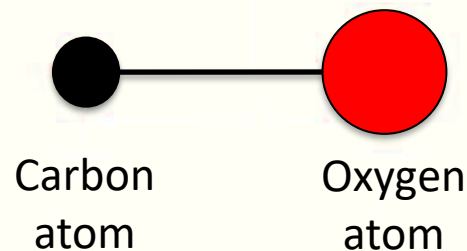
The horsehead nebula in Orion

Observing molecules in space

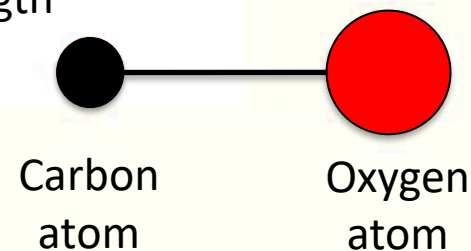
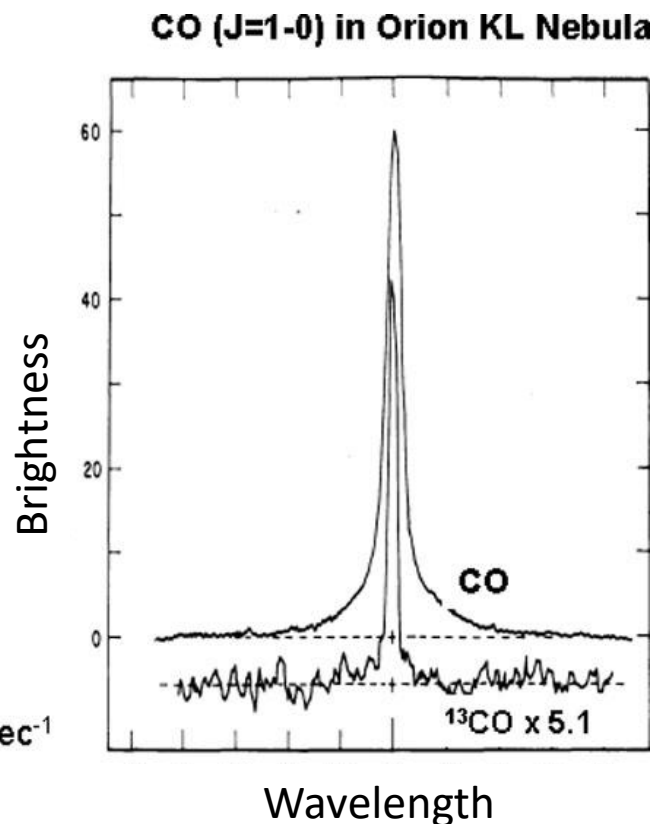
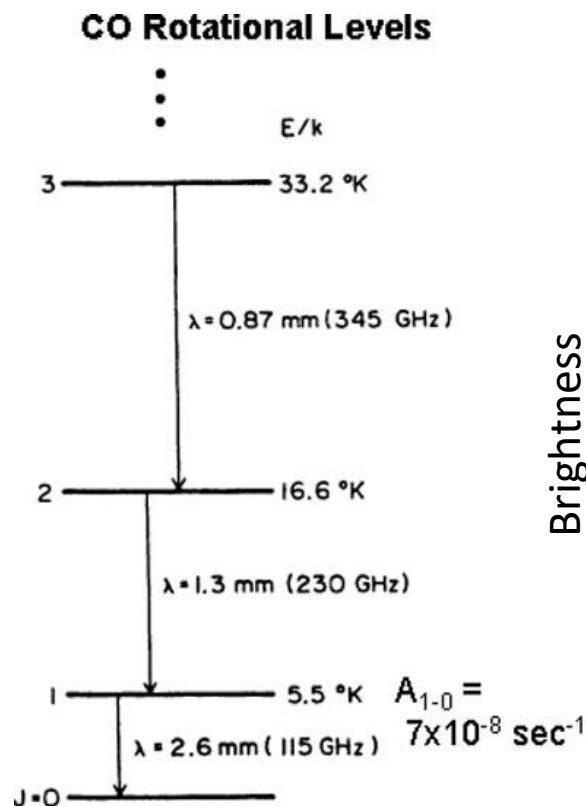


carbon
monoxide

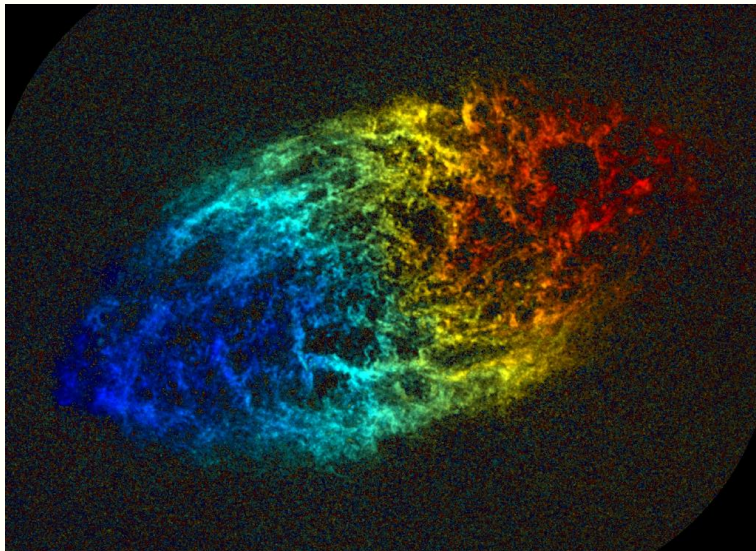
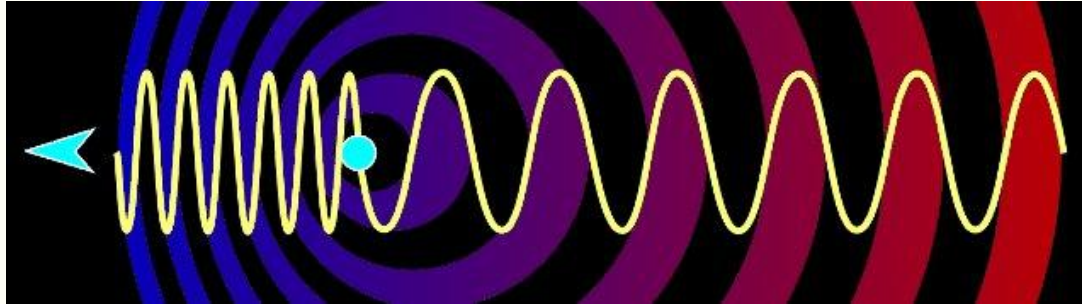
The horsehead nebula in Orion



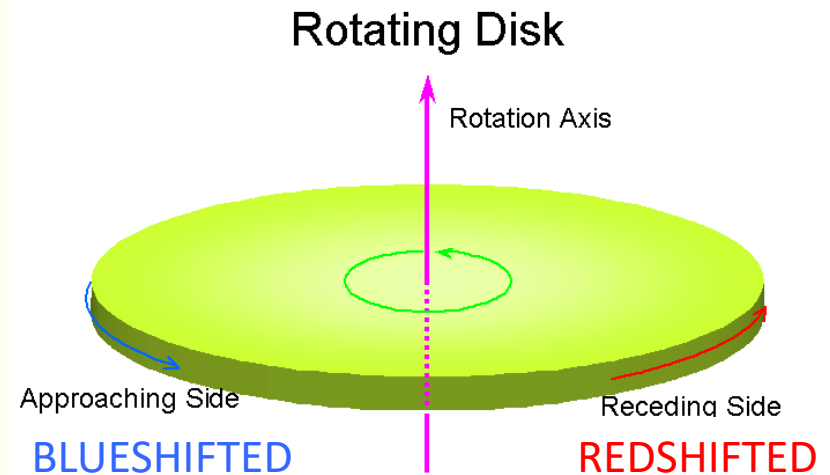
Observing molecules in space

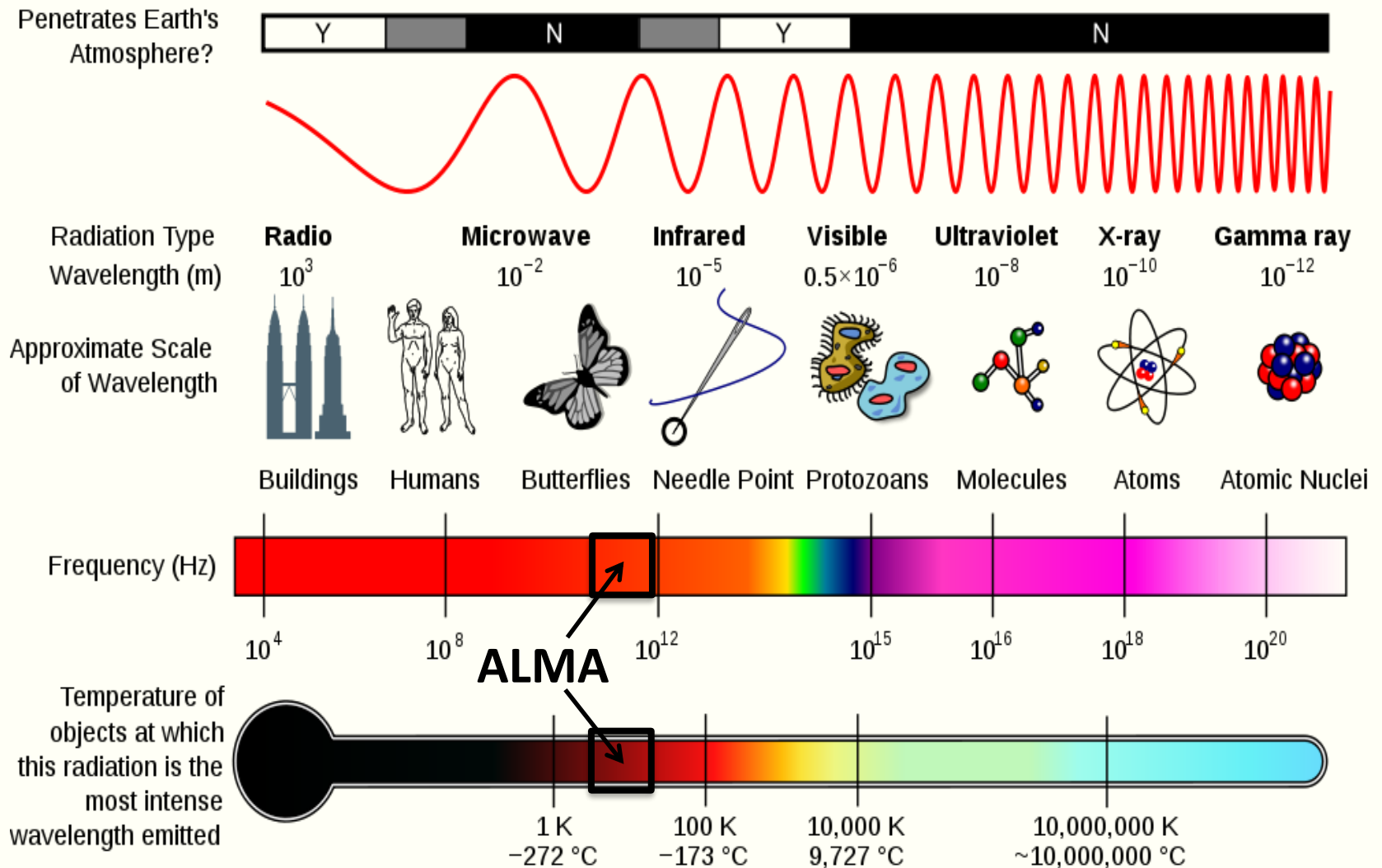


An essential astronomical tool: the Doppler Shift



Velocity field of galaxy M33
Colours show Doppler shift of
line (NRAO, Thilker et al.)







ALMA = **A**tacama **L**arge **M**illimeter/submillimeter **A**rray

Where?

**Chajnantor plateau in the Atacama Desert, Chile
5,050 metres / 16,500 feet above sea level**

What is it?

**66 antennas (or 'dishes') which make
one giant telescope**

The antennas

12m antenna



7m antennas

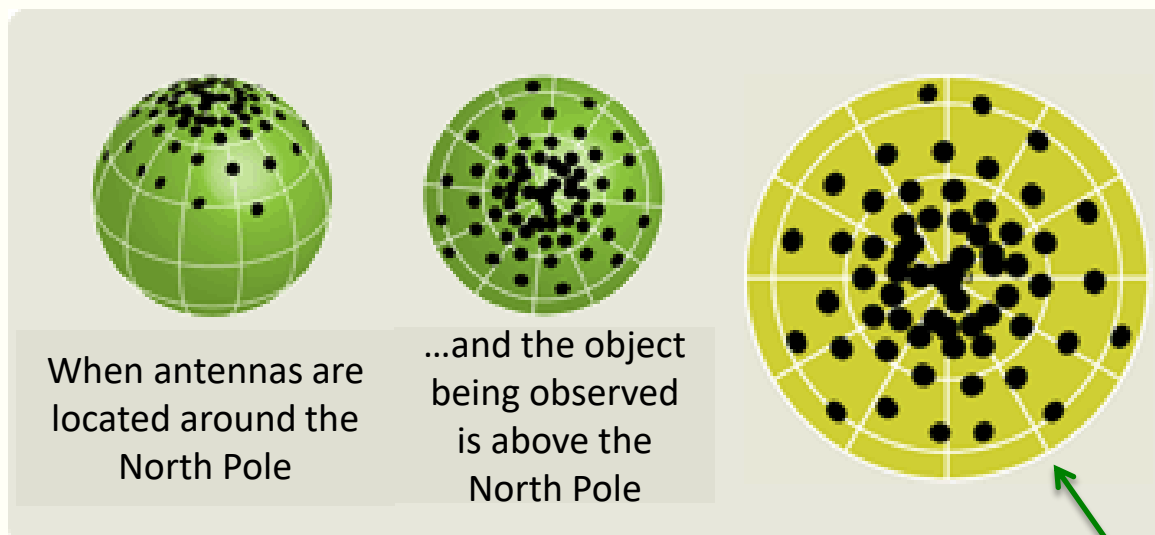


12m antennas



66 dishes make one telescope

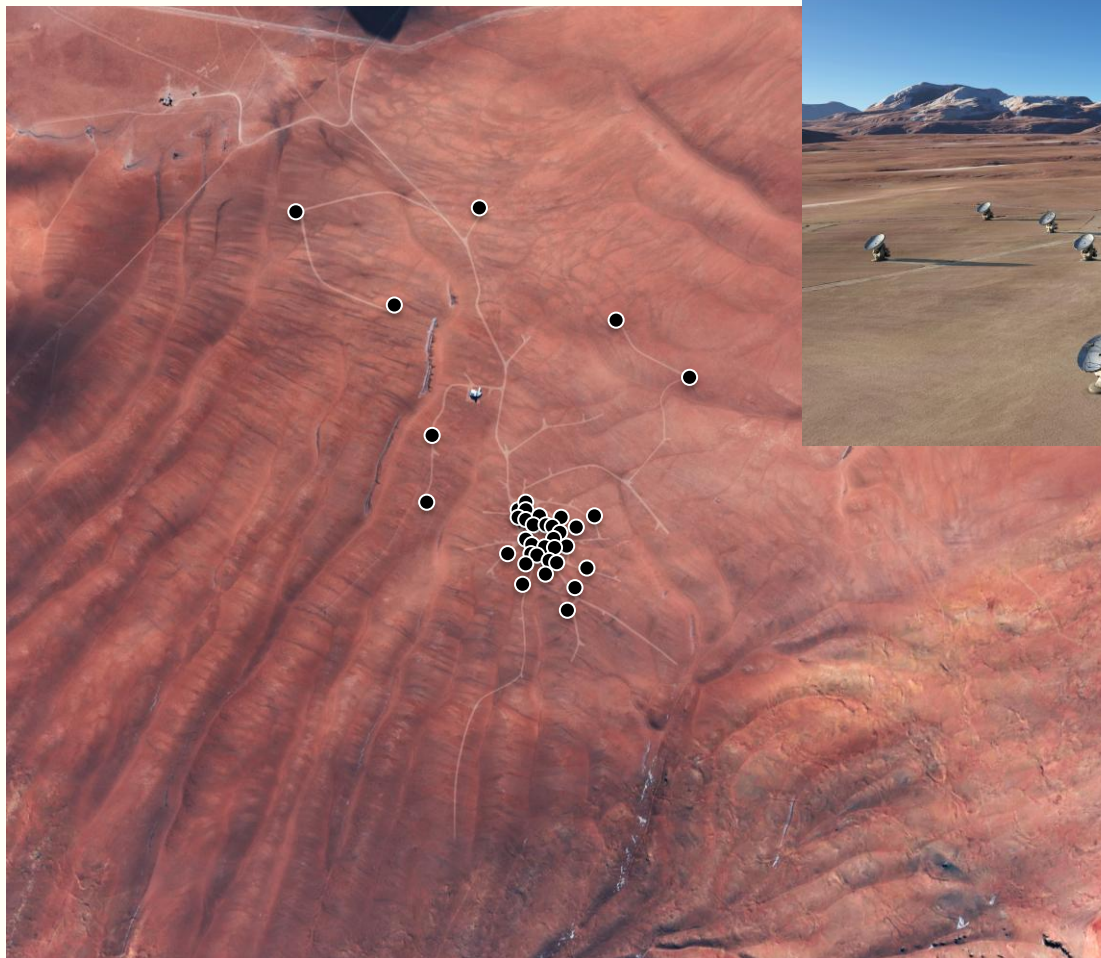
A simple example:



As the Earth rotates
an 'aperture' is made from
from many smaller telescopes

View of Earth
from the object
being observed

ALMA's aperture



● Each dot is
an antenna

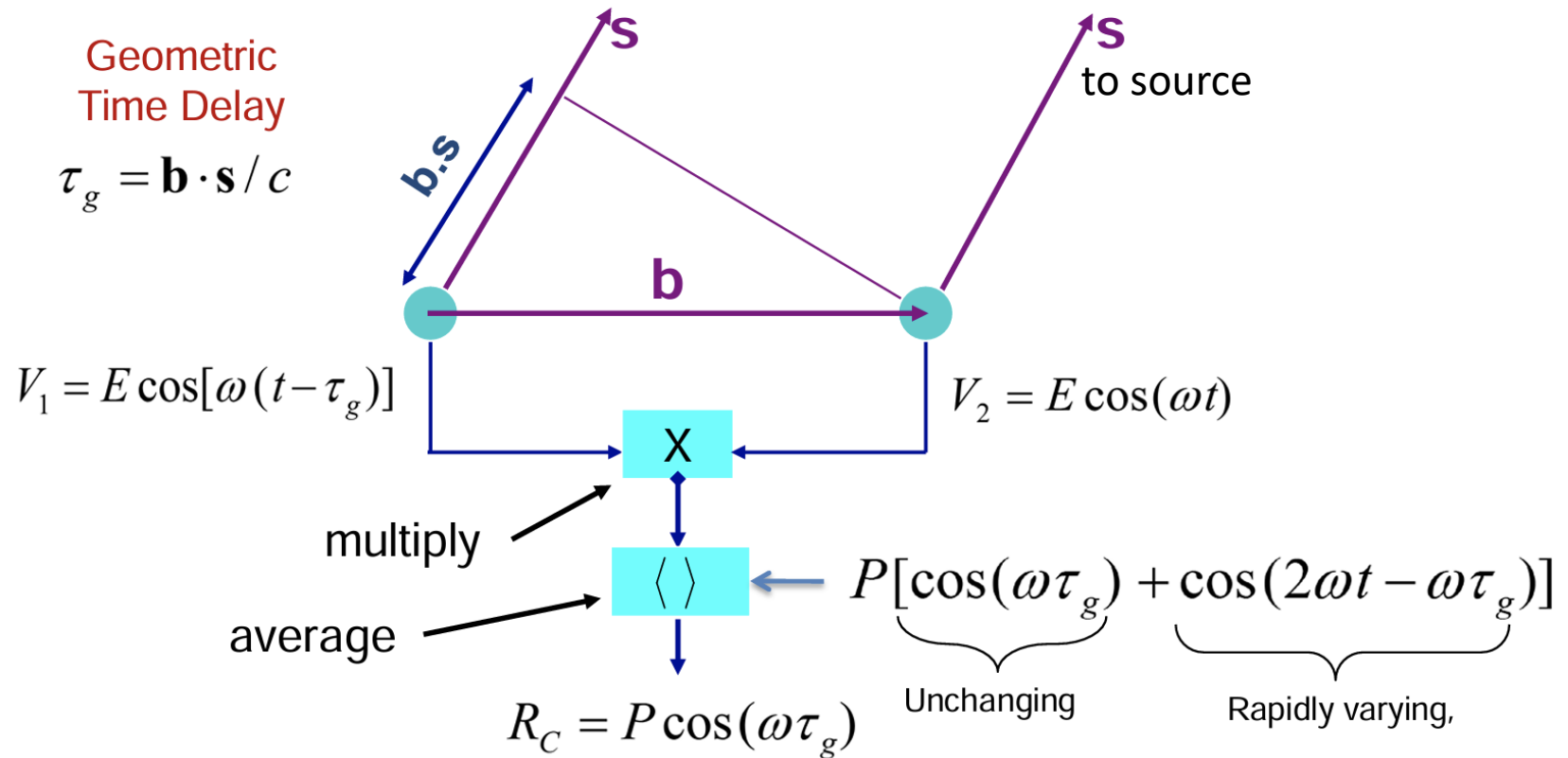
Longest baseline: 16km

Artist impression

Images credit: ALMA (ESO/NAOJ/NRAO)/L. Calçada (ESO)

A basic interferometer (2 antennas)

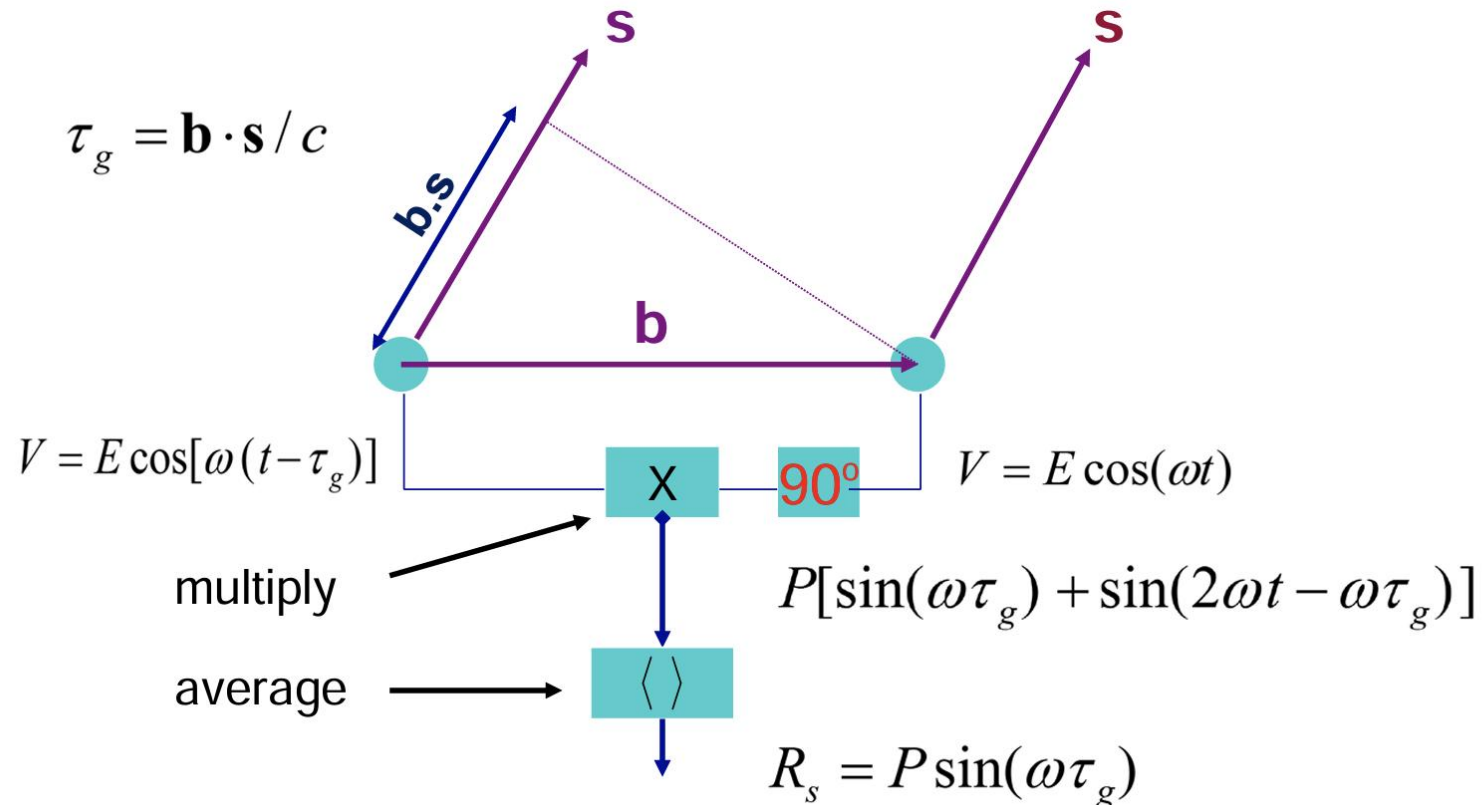
Cosine interferometer



where b is the baseline length, V_1 and V_2 are voltages, $P = E^2/2$ is the power in J s^{-1} and R_C is the correlated signal (after multiplication and averaging)

A basic interferometer (2 antennas)

Sine interferometer



The brightness or intensity $I_\nu(s)$

The brightness or specific intensity $I_\nu(s)$ at a direction on the sky s at a given frequency ν is

$$I_\nu(s) = \frac{P_\nu}{A_{\text{eff}} d\Omega}$$

where A_{eff} is the effective area of the antennas and $d\Omega$ is the area on the sky.

Therefore,

$$R_c = A_{\text{eff}} \int I_\nu(s) \cos(\omega\tau_g) d\Omega$$

and

$$R_s = A_{\text{eff}} \int I_\nu(s) \sin(\omega\tau_g) d\Omega$$

The complex visibility V

Using the complex conjugate of Euler's formula,

$$\cos(\phi) - i\sin(\phi) = Ae^{-i\phi}$$

can define the complex visibility V :

$$V = R_c - iR_s = A_{\text{eff}} \int I_{\nu}(s) e^{-i\omega\tau_g} d\Omega$$

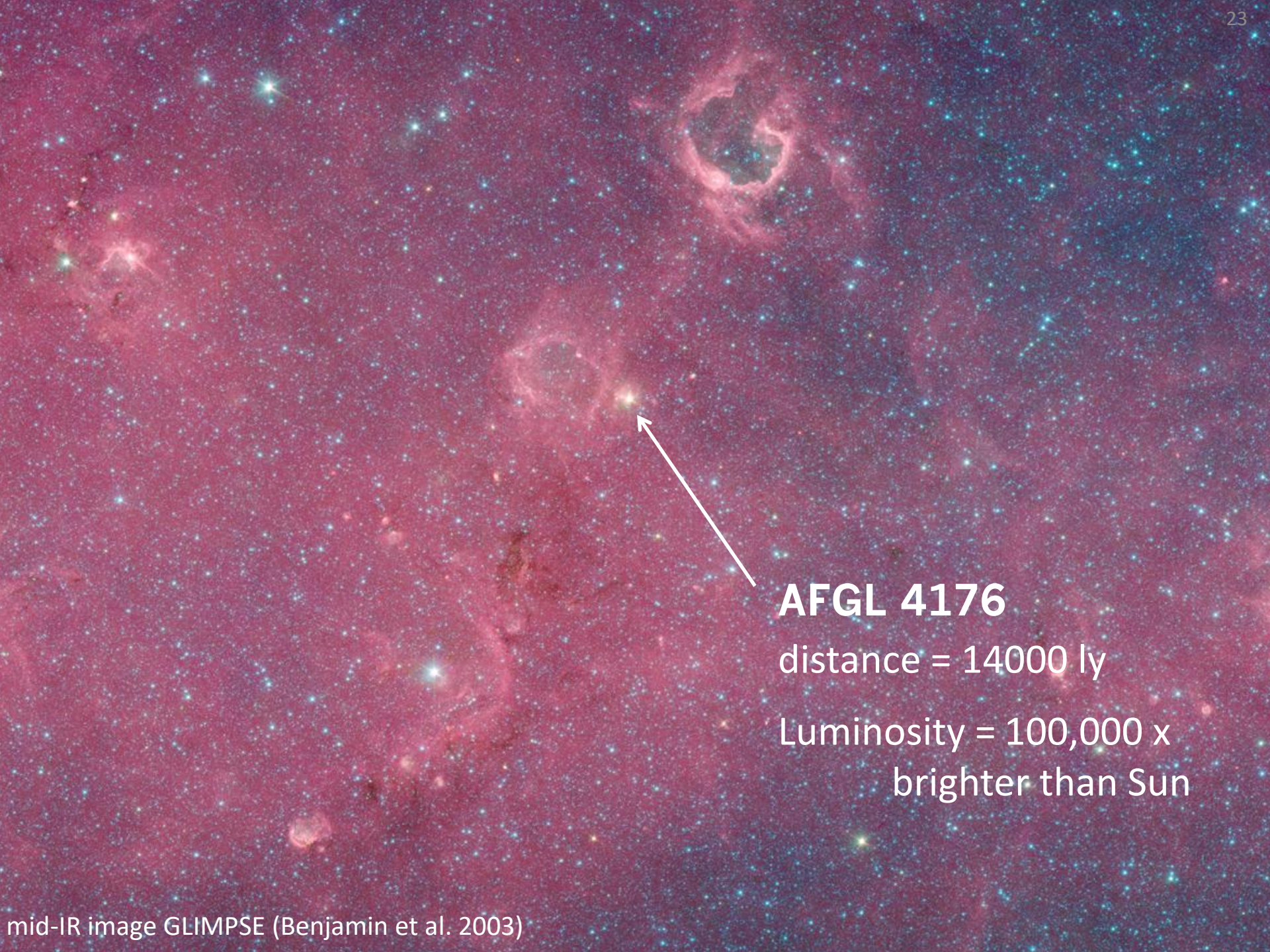
This allows us to relate the correlated signals to the distribution of brightness on the sky $I_{\nu}(s)$. Note that the **RHS in red** is a Fourier transform.

Disks around massive stars with ALMA

Observations of AFGL 4176



Image credit: Clem & Adri Bacri-Normier (wingsforscience.com)/ESO



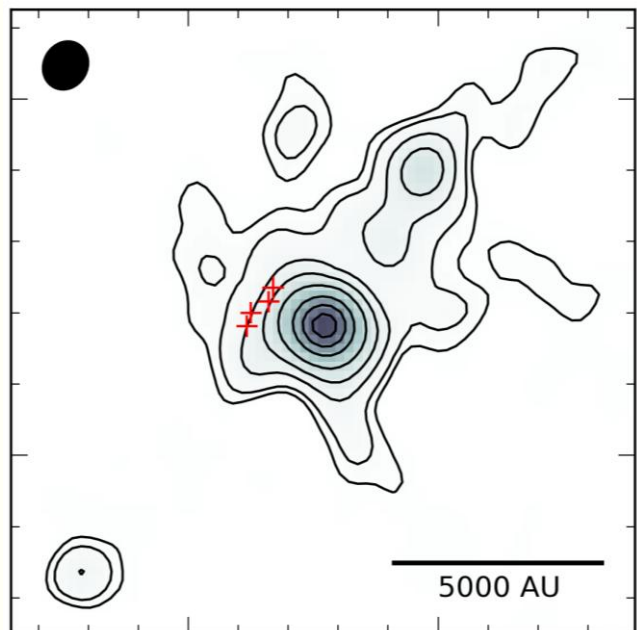
AFGL 4176

distance = 14000 ly

Luminosity = 100,000 x
brighter than Sun

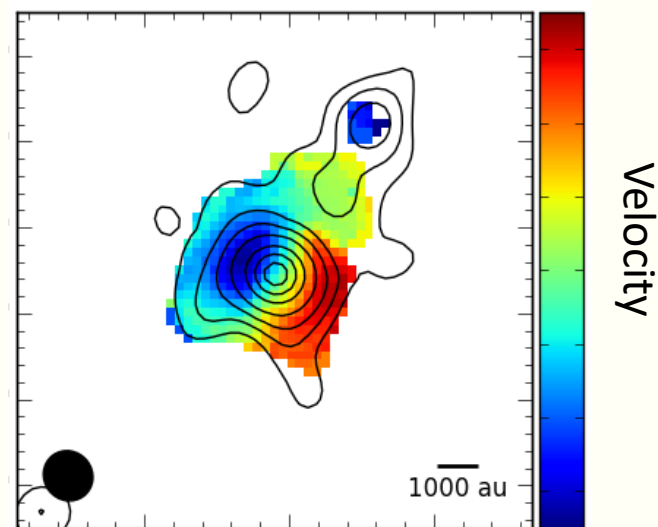
Disks around forming massive stars with ALMA

Emission from dust in disk



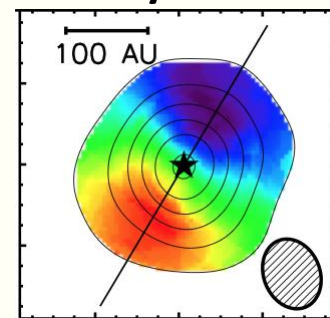
Contours and greyscale: 1.2mm continuum
 Masers shown by +
 Beam (top left): 0.24 x 0.28 arcseconds

Emission from the molecule CH_3CN



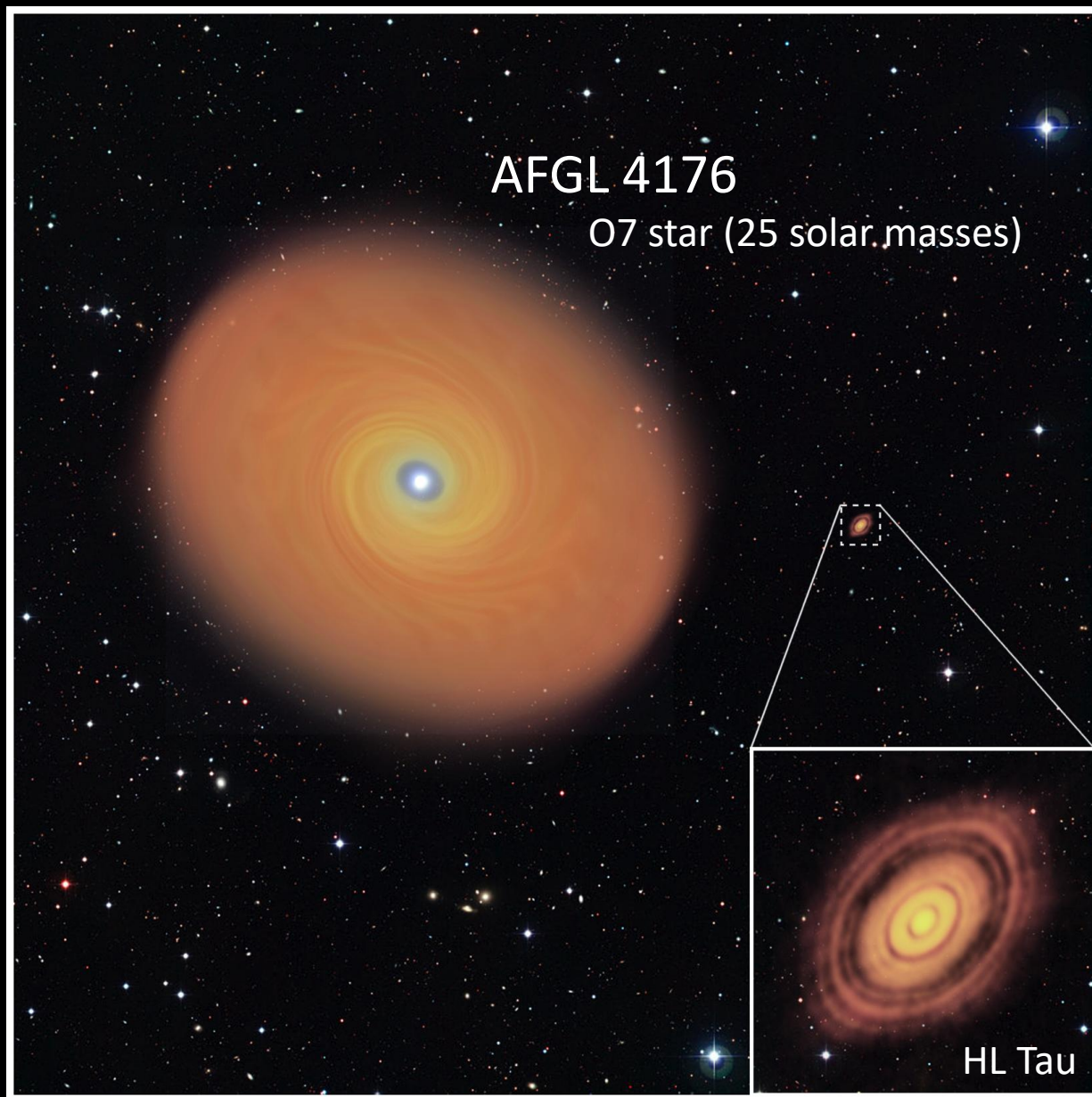
Johnston et al. (2015)

TW Hya:



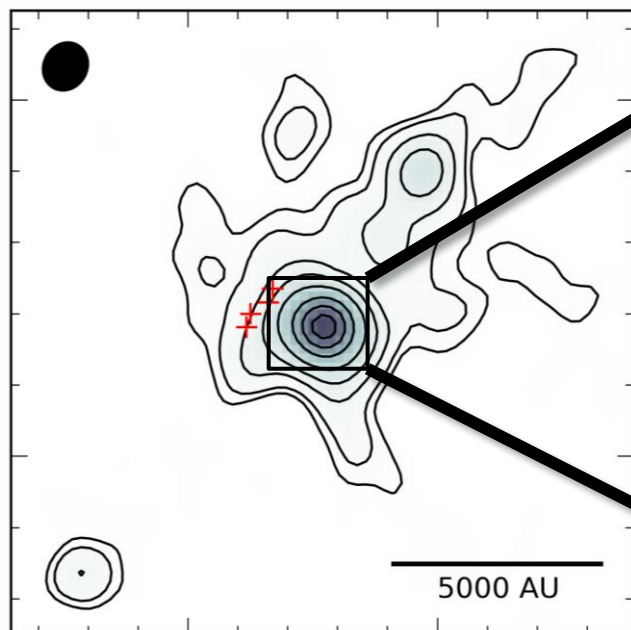
Comparison
 to low mass
 protostar:

Hughes et al. (2011)



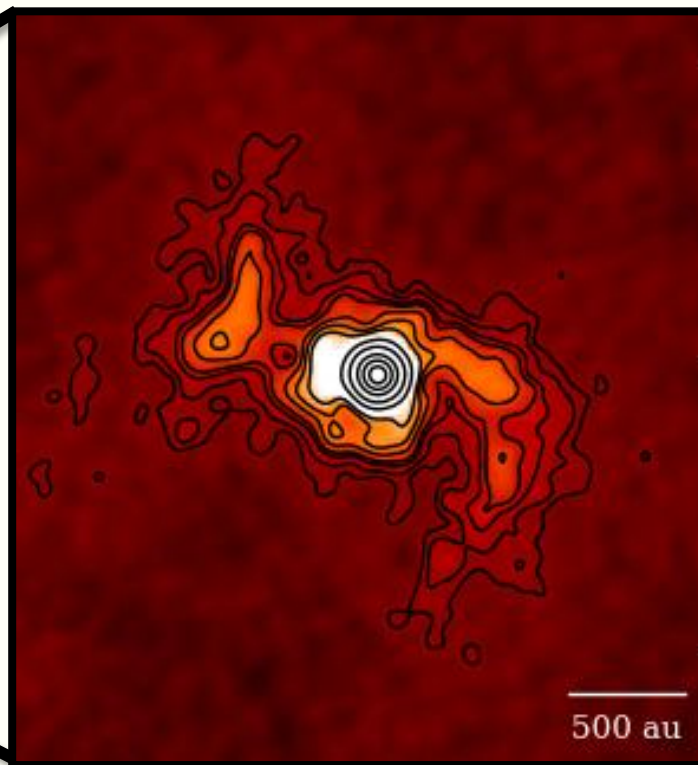
Higher-resolution image with ALMA

Emission from dust in disk



Longest baseline: 1km

Emission from dust in disk



Johnston et al. (2020)

Longest baseline: 12.3km

Other big telescopes / interferometers

The Very Large Array (VLA)
max baseline 36 km



The Square Kilometer Array (SKA)
Max baselines 150 and 74km



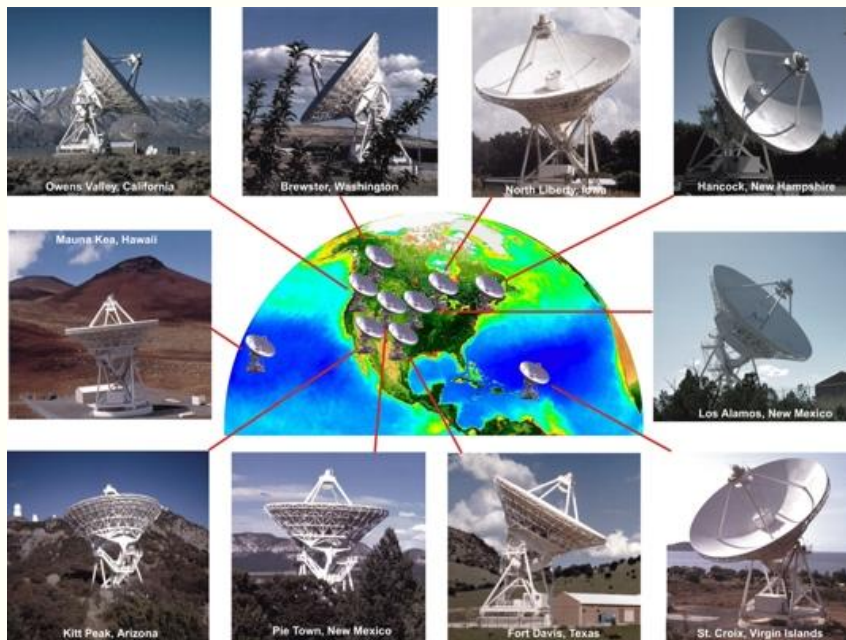
SKA-Mid

SKA-Low



e-MERLIN (Multi-Element Radio Linked
Interferometer Network)
Max baseline: 217 km

Very Long Baseline Interferometry



Very Long Baseline Array (VLBA)

What is e-VLBI?

Example: the **European VLBI Network (EVN)**

- Signals are sent and correlated in real time (same for “regular” interferometry)
- Different types of radio interferometry not very different in the end (mostly due to data collection and calibration methods)

European VLBI Network (EVN) – max baseline 10000 km

Very Long Baseline Array (VLBA)

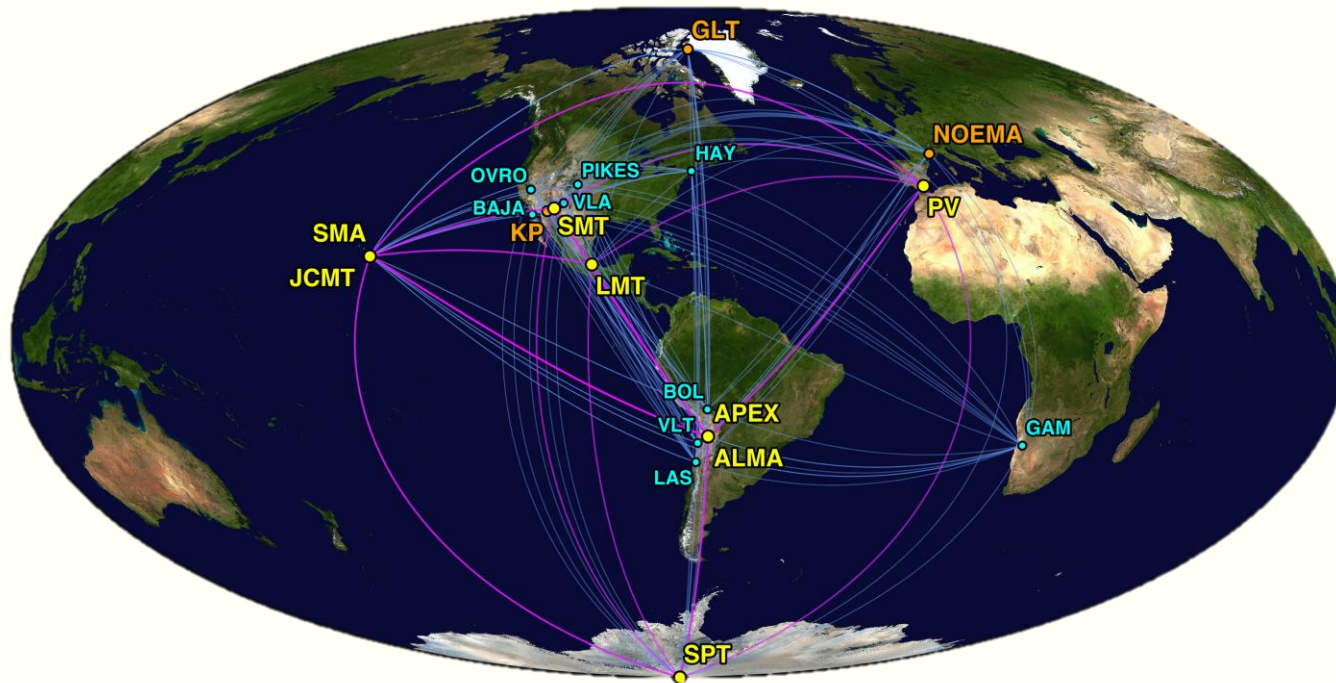
Max baseline: 8611 km!

Differences between regular and **Very Long Baseline Interferometry (VLBI)**:

- Signals are not correlated (multiplied and averaged) in real time
- Collected data is timestamped using atomic clocks or radio timing signals
- Data sent later (by internet or hard drive) to correlator at one location
- Excellent to detect very compact, bright objects



Event Horizon telescope



- One of the World's largest Interferometers (maximum baseline: 10700 km)
- Combines many different telescope facilities from across the world (including ALMA)
- Wavelength of observation: 1.3mm and recently 0.87mm
- Can therefore achieve a resolution (image detail) of **0.000019 arcseconds**
- Technique is VLBI: data is stored on disks and shipped to supercomputers at MIT and Max Planck Institute for Radio Astronomy in Germany

Black hole at Centre of M87

- EHT observed supermassive black hole at the centre of M87 (also called M87*)
- Resolved close to the radius of the Event Horizon, also called the Schwarzschild radius (the radius relating to the escape velocity for light):

$$r_s = \frac{2GM}{c^2}$$

- The mass of M87* is 6.5 billion solar masses(!)
- Therefore, $r_s = 128$ au (au = Earth-Sun distance)
- We can find the size of the “Schwarzschild diameter” on the sky using:

$$\theta(\text{arcsecond or ''}) = \frac{D(\text{au})}{d(\text{parsec})} = 0.000015'' \text{ or } 15 \mu\text{as}$$

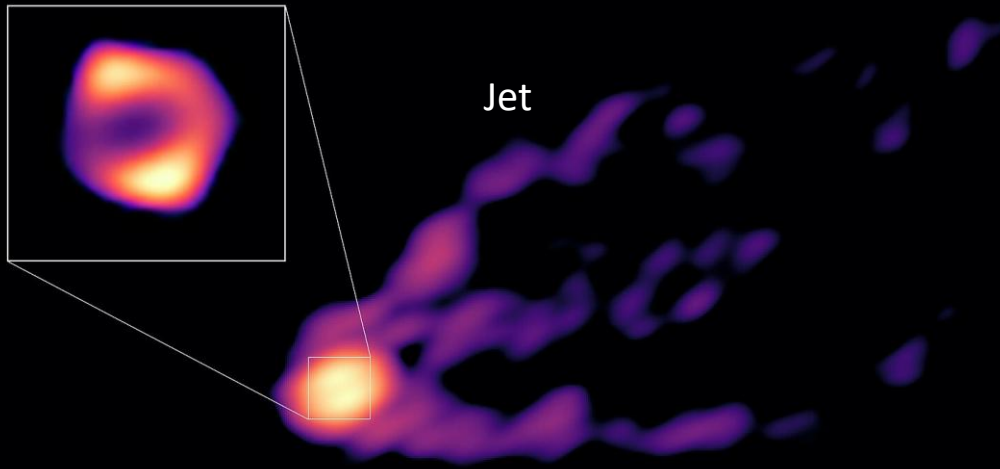
Where θ is the size on the sky, D is the physical size and d is the distance

M87*

- EHT results were announced in 2019
- Resolved the emission from material orbiting the black hole...
- ...due to relativistic acceleration of electrons in a magnetic field (synchrotron emission)

GMVA+ALMA+GLT image, wavelength: 3.5mm

Ring diameter: $64 \mu\text{as}$

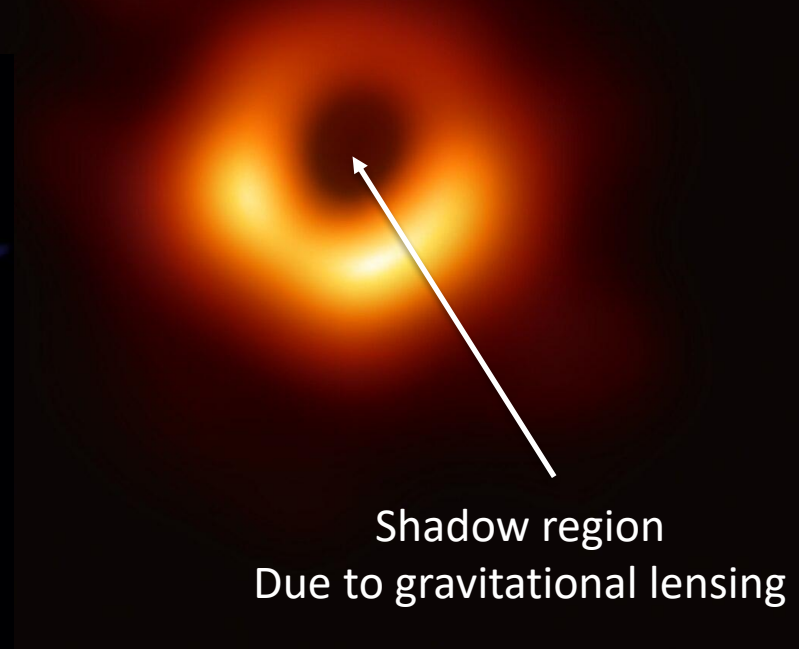


Jet

Image taken with the Global Millimetre VLBI Array (GMVA), ALMA and the Greenland Telescope (GLT)

EHT image, wavelength: 1.3mm

Ring diameter: $42 \mu\text{as}$

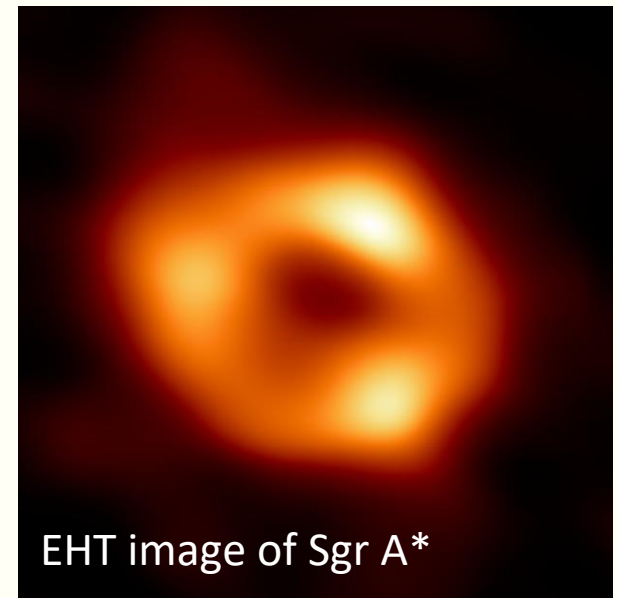


Shadow region
Due to gravitational lensing

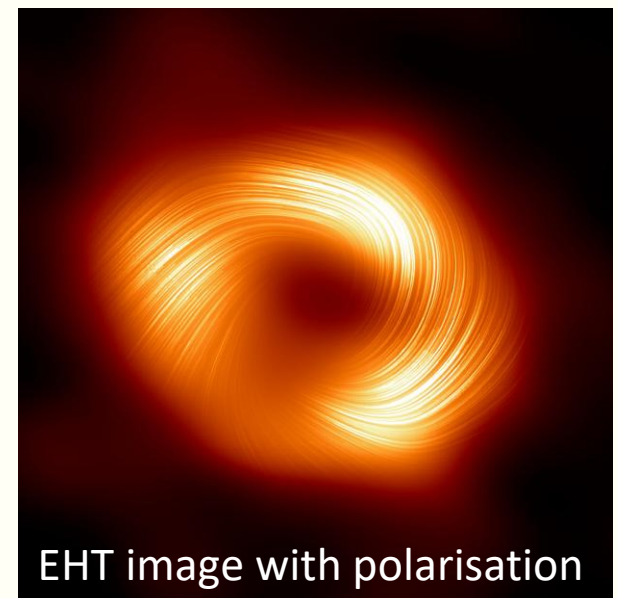
EHT image of material around super-massive black hole M87*

Sgr A*

- Results announced in 2022
- Sgr A* is the black hole at the centre of our Galaxy, the Milky Way
- Ring has diameter of $52 \mu\text{as}$
- However, M87* has 1500 more mass and is 2000 times further away!
- Can deduce: lens rings are common features of black holes
- The polarisation of emission in M87* and Sgr A* was measured and modelled to determine:
 - the angular momentum of the black hole
 - that the magnetic field is dynamically important

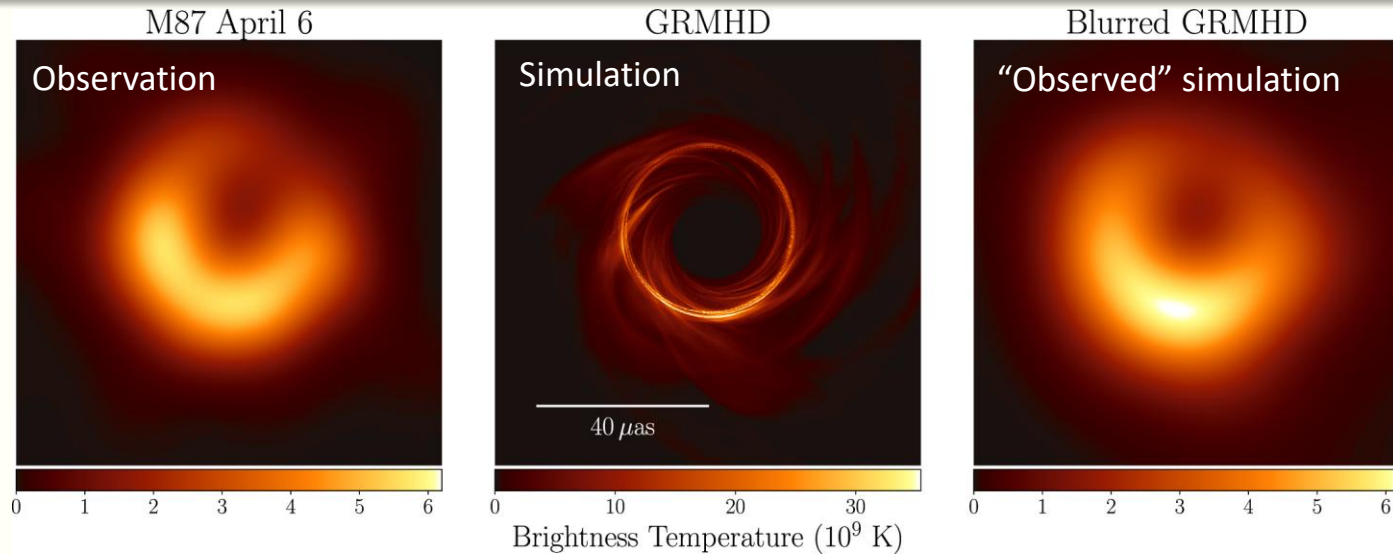


EHT image of Sgr A*

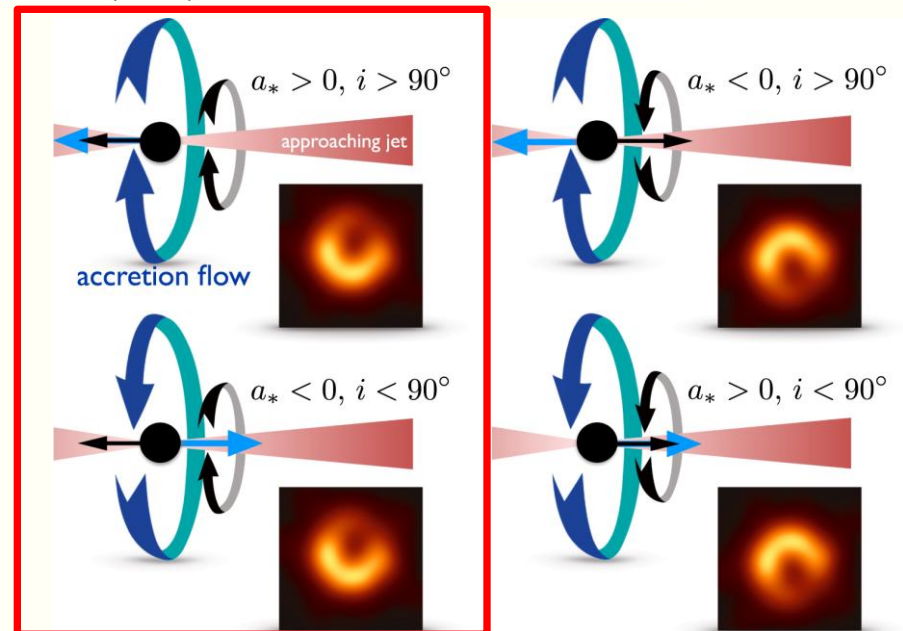


EHT image with polarisation

Comparison with theory (M87*)



- General Relativity Magneto-hydrodynamic (GRMHD) Simulations to model the observations
- Model produces gravitational lensing of synchrotron emission from hot plasma
- Modelling shows the black hole has to be spinning clockwise (as viewed from earth)



Blue-green ribbon: accretion disk; black ribbon: black hole

Summary

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- Peering through interstellar dust using radio observations
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- Even bigger interferometers and the EHT
- Science results from the EHT: observations of supermassive black holes